

Recursion and Immutable Data

CS 350

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THESE ARE YOUR
FATHER'S PARENTHESSES



ELEGANT
WEAPONS



FOR A MORE... CIVILIZED AGE.

Overview

Objectives

- To learn how to use Dr. Racket

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- To learn how to install and use `#lang flit`

Programming in CS 350

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- The Dr. Racket editor

Racket

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- Language for making languages
 - You will find documentation for many of these languages online
 - They are different from what we are doing. Beware!

What is Dr. Racket?

- IDE for Racket

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 - Feedback when writing code
 - Can evaluate expressions while you're writing your code

Flit

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 - You can do a lot with very little

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- Data Types with Variants

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 - NOT always a named function!

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- Languages like C++ and Python use “infix notation” for math

Function calls

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- In Flit, you write `( f x y z )`

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- In C++ you write `f(x, y, z)`
- In Flit, you write `(f x y z)`
- 99% of what you do in Flit is calling functions

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 - Hierarchical structure: expressions contain expressions contain expressions...
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- JSON is basically just funny-looking parentheses

Example

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(+ 2 7)
(- 10 0.5)
(* 1/3 2/3)
(/ 1 100000000000000.0)
(max 10 20)
(modulo 10 3)
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9.5
2/9
1e-12
20
1
```

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(= (+ 2 3) 5)
(> (/ 0 1) 1)
(zero? (- (+ 1 2) (+ 3 0)))
(and (< 1 2) (> 1 0))
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 - *Mutate* value of variable

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- Use functions and recursion to do all computation

Example: Imperative Conditionals

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```
if (someCondition)
  doThis
else
  doThat
```

- The if-statement doesn't have a value, but it can mutate (change) the state

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  (if (= 2 (+ 1 1))  
    3  
    40))
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 - e.g. Below, the 2nd if evaluates to 3 because condition is true

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```
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```


Functions

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(define (isRemainder [x : Number]
                  [y : Number]
                  [remainder : Number])
  : Boolean
  (= remainder (modulo x y)))
(isRemainder 10 3 1)
(isRemainder 10 4 1)
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```
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- General form:

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Functions (ctd.)

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```
(define (functionName  
    [argName : argType]  
    ...  
    [argNameN : argTypeN]) : returnType  
  functionBody)
```

- Later in the course we'll see another way of defining functions

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 - Compares pointers, so very fast

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  (let ([xy (+ x y)])
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 - We'll see this later when we learn about lambdas

Functional Thinking and Recursion

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 - Fast, memory efficient

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- Programming is **declarative**

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Further reference:

<http://htdp.org>, Matthew Flatt's Notes (URCourses)

Factorial - Representation

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```
(define (factorial [n : Number]) : Number  
  (error 'factorial "TODO"))
```

Factorial - Examples

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```
(test (factorial 0) 1 )  
(test (factorial 1) 1 )  
(test (factorial 2) 2 )  
(test (factorial 3) 6 )  
(test (factorial 4) 24 )  
(test (factorial 5) 120 )
```

- Notice the pattern?

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 - Zero

Factorial - Template

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```
(define (factorial [n : Number]) : Number
  (if (zero? n)
    (error 'zero "TODO")
    (let ([n-1 (- n 1)])
      (error 'suc "TODO")))
  ))
```


Factorial - Recursion

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```
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      (let*
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           [fn-1 (factorial n-1)])
        (error 'suc "TODO")))
  ))
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Factorial - Filling holes

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```
(define (factorial [n : Number]) : Number
  (if (zero? n)
    1
    (let*
      ([n-1 (- n 1)]
       [fn-1 (factorial n-1)])
      (* n fn-1))
  ))
```


Run Tests

```
(test (factorial 0) 1 )  
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```
good (factorial 0) at line 11  
  expected: 1  
  given: 1
```

```
good (factorial 5) at line 12  
  expected: 120  
  given: 120
```

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 - Aside: I research languages where you *can* express this with types

Another Example: Exponentiation

- Live coding in Dr. Racket

Unbounded Data: Lists

Functional Linked Lists

- Every linked list is one of:

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 - Creates a new list whose tail is the old list

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- Lots more helper functions, see the documentation

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```
(define (list-template
  [xs : (Listof Number)])
  (if (empty? xs)
      (error 'nil "TODO")
      (let ([h (first xs)]
            [t (rest xs)])
        [tRet (list-template t)])
      (error 'cons "TODO")))
))
```


Example: Sum

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```
(define (sum [xs : (Listof Number)])  
  : Number  
  (if (empty? xs)  
      0  
      (let* ([h (first xs)]  
              [t (rest xs)]  
              [tRet (sum t)])  
        (+ h tRet))))  
(sum '())  
(sum '(1 2 3))  
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```

0

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 - e.g. `first '()` will raise an error

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```
(define (increment [xs : (Listof Number)])  
  : (Listof Number)  
  (type-case (Listof Number) xs  
    [empty  
     empty]  
    [(cons h t)  
     (cons (+ h 1) (increment t))]))  
(increment '(2 3 4))
```

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```
'(3 4 5)
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 - e.g. `first '(1 2 3)` is a Number, but `first '(#t #f #t)` is a Boolean
- Later, this will be very useful for writing generic list operations

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```
(define (concat [xs : (Listof 'elem)]  
            [ys : (Listof 'elem)])  
  : (Listof 'elem)  
  (type-case (Listof 'elem) xs  
    [empty  
     ys]  
    [(cons h t)  
     (cons h (concat t ys))]))
```

Example: List Concatenation (ctd.)

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Example

```
(concat '(1 2 3) '(4 5 6))  
(concat '("3" "5") '("0"))  
(concat '() '(#t))  
(concat '(#f) '())
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Example

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(concat '(1 2 3) '(4 5 6))  
(concat '("3" "5") '("0"))  
(concat '() '(#t))  
(concat '(#f) '())
```

Results

Example: List Concatenation (ctd.)

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Results

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More Examples

- Demo: Dr. Racket (as time permits)

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 - Filtering out odd elements of a list